Extremely intense neutron sources can be constructed from a high energy ($\sim$1 GeV) high current proton linear accelerator and a heavy element (e.g. molten Pb) spallation target. Tentative designs have been described that would be capable of generating $\sim$0.1 g of neutrons (0.1 mole) per second. Such intense neutron sources could be used to transmute longlived radioactive nuclides (into inactive or shorter lived ones) or drive a subcritical assembly for production of energy by fission of actinides.

16.9. AREAS OF APPLICATION FOR ACCELERATORS

Let us consider a few uses of accelerators. In §16.8 we have discussed the use of accelerators for neutron production. Electron as well as positive particle beams from these accelerators can be used directly for the study of radiation effects in materials (see Ch. 8), to produce X-rays, or to generate intense light. Accelerators are used in medicine for producing intense γ- or X-ray beams for cancer treatment, or charged particle beams for similar purposes (the "proton-knife"). Accelerators in the 10 MeV range are very useful in producing radionuclides, which either cannot be made through (n,γ)-reactions in nuclear reactors or are very shortlived. Small cyclotrons of a size suitable for use in hospitals are commercially available. The precise definition of energy available in van de Graaff accelerators makes them particularly useful for the study of the energy levels of nucleons in nuclides and for ^{14}C dating. VdG's, cyclotrons, linear accelerators, and synchrotrons have been extensively used for the study of nuclear reactions.

Very high energy accelerators are used for the production and study of elementary particles. Once the beam energy exceeds a couple of hundred MeV, mesons and other elementary particles are produced in nuclear reactions and can be studied directly or they can be used to cause various types of nuclear reactions.

16.9.1. Syncrotron light sources

High energy electron accelerators, mostly in form of large diameter storage rings, are used to produce high intensity synchrotron radiation for basic studies in chemistry, material science and biology. The synchrotron "light" is usually obtained by placing bending magnets in the path of the circulating electron beam. A series of opposing strong magnets repeatedly bend the beam back and forth many times. These devices are characterized by their K_U-value which is defined as given below.

$$K_U = \frac{eB\lambda}{2\pi v m_e} \tag{16.14}$$

where v is the electron velocity in the beam and λ is the distance beween magnets with the same orientation. When K_U is $\ll 1$ the device is called an undulator and produces x-rays in a narrow energy band, but when K_U is $\gg 1$ the device is called a wiggler and produces a continuum of x-rays within a rather wide energy range. A typical device is

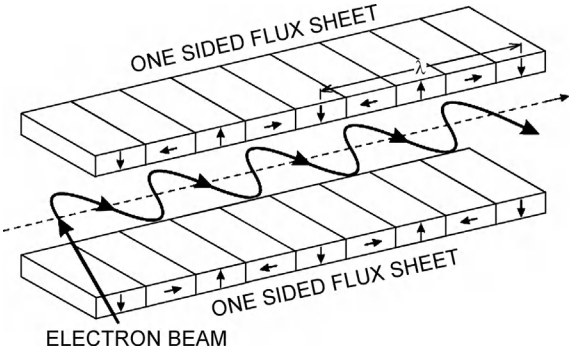


Figure 16.12 Principle of undulators and wigglers for production of intense synchrotron radiation. The X-rays are emitted in the direction shown by the dashed line with an arrow at the right.

shown schematically in Figure 16.12. The direction changes will increase the intensity of the x-ray beam due to the large number of bendings. Syncrotron light sources are used for different kinds of spectroscopy such as, e.g., EXAFS (Extended X-ray Absorption Fine Structure) which has grown considerably in popularity with chemists in the last decade due to the ability to directly study complexation in solution and solids.

16.10. EXERCISES

16.1. In a small linear accelerator containing 30 stages, He^{2+} ions are accelerated by a 150 kV, 100 MHz RF source. The ions are used to bombard a metal target to induce a specific reaction. (a) What is the proper length of the last drift tube? (b) What is the maximum projectile energy achieved? (c) What is the heaviest target in which a nuclear transformation can be induced (no tunnelling)?

16.2. Assume a linear accelerator is built in three sections, each with 30 stages of 100 kV. Between which sections should a thin carbon stripper foil be placed in order to achieve the highest final energy, when accelerating ^{84}Kr ions from an ion source emitting low energy $^{84}Kr^{4+}$ ions? Consider the nearest integer charge after stripping.

16.3. In order to propel a space vehicle to high velocities after its exit from the earth's gravitational pull, "ion rockets" might be used. These can be considered as simple accelerators for charged particles. The electroneutrality of the vehicle is conserved through emission of electrons from a hot filament. The gain in linear velocity Δv is given by the "rocket formula":

$$\Delta v = v_e \ln(m_{Ro}/m_R)$$

where v_e is the exhaust velocity and m_{Ro} and m_R are the initial and final mass of the vehicle respectively. (a) Calculate the propelling power for a rocket, which emits 10 keV protons at a current of 1 A. (b) What is the final velocity gain of such vehicle with 2000 kg initial mass after 1 year's operating time? (c) For the same net available power and operating time, as in (b), calculate the final velocity gain of a 2000 kg vehicle which emits 10 keV Cs^+ ions at a current of 1 A.